INTRODUCTION TO THE FINITE ELEMENT METHOD FOR ELLIPTIC PROBLEMS

- Variational weak formulation for a 1D model problem
- FEM for the model problem using piecewise linear functions
- Error estimation for the 1D model problem
- FEM for the 2D Poisson equation
- Introduction to Hilbert spaces $L_2(\Omega)$, $H^1(\Omega)$ and $H_0^1(\Omega)$
- The best error approximation: Geometric interpretation
- The Neumann problem: Natural and essential boundary conditions
- FEM Programming

INTRODUCTION TO FEM FOR ELLIPTIC PROBLEMS

Variational formulation of a one-dimensional model problem

$$\boxed{\begin{gathered} -u''(x) = f(x), \quad 0 < x < 1 \\ u(0) = u(1) = 0 \end{gathered}}$$

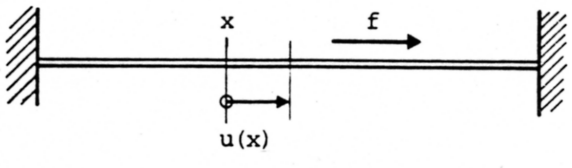
THE STRONG OR DIFFERENTIAL PROBLEM (D)

- $u' = \frac{du(x)}{dx}$, and $f(x)$ is a given continuous function.
- By integrating twice, we can see that this problem has a unique solution.

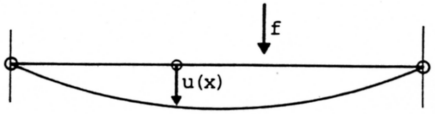
SOME EXAMPLE PROBLEMS

- An elastic bar

$\sigma = Eu'$	(Hooke's law)
$\sigma' = -f,$	(Equilibrium equ.)
$u(0) = u(1) = 0$	(Boundary conds.)



- An elastic cord



- Heat conduction

$q = -Ku'$	(Fourier's law)
$q' = f,$	(Conservation of energy)
$u(0) = u(1) = 0$	(Temperature boundary conds.)

THE MINIMIZATION PROBLEM (M)

- Define the linear functional $F : V \longrightarrow \Re$ as:
  $F(v) = \frac{1}{2}(v', v') - (f, v)$
- Find $u \in V$ such that $F(u) \quad \leq \quad F(v) \;\; \forall\, v \in V$

where:

- $V = \{v : v$ is a continuous function on $[0, 1]$, v' is piecewise continuous and bounded on $[0, 1]$, and $v(0) = v(1) = 0\}$
- $(v, w) = \int_0^1 v(x)w(x)dx$, for real valued piecewise continuous functions.
- (M) corresponds to the *'Principle of Minimum Potential Energy in Mechanics'*.

THE VARIATIONAL (WEAK) PROBLEM (V)

$$\boxed{\text{Find } u \in V \text{ such that } (u', v') = (f, v) \;\; \forall\, v \in V}$$

where:

- v is the so called test function
- (V) corresponds to the '*Principle of Virtual Work in Mechanics*'.

The solution of (D) is also a solution of (V)

- We multiply the equation $-u'' = f$ by an arbitrary function $v \in V$
- Integrate over $(0, 1)$ which gives:
  $-(u'', v) = (f, v)$
- We now integrate the left-hand side by parts using the fact that $v(0) = v(1) = 0$ to get:
  $-(u'', v) = -u'(1)v(1) + u'(0)v(0) + (u', v') = (u', v')$
- We conclude that: $(u', v') = (f, v) \;\; \forall\, v \in V$, i.e. u is a solution of (V).

The solution of (V) is also a solution of (M)

- Let u be a solution to (V), let $v \in V$ and set $w = v - u$ so that $v = u + w$, $w \in V$.
- We can write:
$$\begin{aligned} F(v) = F(u+w) &= \tfrac{1}{2}(u'+w', u'+w') - (f, u+w) \\ &= \tfrac{1}{2}(u', u') - (f, u) + (u', w') - (f, w) + \tfrac{1}{2}(w', w') \\ &= F(u) \qquad + \qquad 0 \qquad + \tfrac{1}{2}(w', w') \\ &\geq F(u) \end{aligned}$$
- Thus, u is the minimizer of $F(v)$ (u is a solution of (M)).

The solution of (M) is also a solution of (V)

- Let u be a solution to (M), then for any real number ϵ and any $v \in V$ we have that $u + \epsilon v \in V$, which implies:
$$F(u) \leq F(u + \epsilon v)$$
- Thus the differentiable function $g(\epsilon) = F(u + \epsilon v)$ has a minimum at $\epsilon = 0$ and hence $g'(0) = 0$.
- We have:
$g(\epsilon) = \frac{1}{2}(u', u') + \epsilon(u', v') + \frac{\epsilon^2}{2}(v', v') - (f, u) - \epsilon(f, v)$ and
- $g'(\epsilon) = (u', v') + \epsilon(v', v') - (f, v)$.
- Using $g'(0) = 0$ and $g'(0) = (u', v') - (f, v)$, results in:
- $(u', v') = (f, v)$, i.e. u is a solution of (V).

Problem (V) has a unique solution

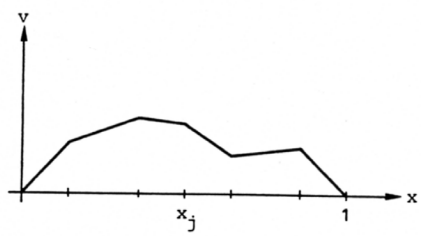
- Let (V) admit two solutions $u_1, u_2 \in V$.
- $(u_1', v') = (f, v) \;\; \forall\, v \in V$ and
  $(u_2', v') = (f, v) \;\; \forall\, v \in V$
- Subtracting these equations leads to the following:
  $\int_0^1 (u_1' - u_2') v' dx = 0 \;\; \forall\, v \in V$.
- Choosing $v = u_1 - u_2 \in V$ results to:
  $\int_0^1 (u_1' - u_2')^2 dx = 0$, which shows that:
- $u_1'(x) - u_2'(x) = 0 \;\; \forall\, x \in [0, 1]$.
- It follows that $u_1(x) - u_2(x) = \mathit{constant}$ on $[0, 1]$, which together with the boundary condition $u_1(0) = u_2(0) = 0$ gives that:
  $u_1(x) = u_2(x) \;\; \forall\, x \in [0, 1]$, and the uniqueness is proved.

If u is a solution of (V) then u also satisfies (D)

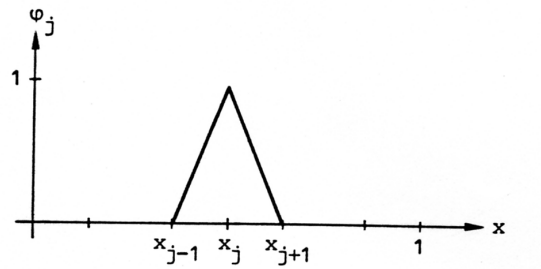
- Let $u \in V$ be a solution of (V), then:
$$\int_0^1 u'v'dx - \int_0^1 fvdx = 0 \;\; \forall\, v \in V \qquad (*)$$
- We assume that u'' exists and is continuous (**regularity condition**).
- Integration of the first term in Equ. (*), and using $v(0) = v(1) = 0$, leads to:
$$\int_0^1 (u'' + f)vdx = 0 \;\; \forall\, v \in V$$
- By the assumption that $(u'' + f)$ is continuous, the above relation can only hold if:
$$u''(x) + f(x) = 0,\, x \in [0, 1]$$
- So u is a solution of (D).

FEM for the model problem with piecewise linear functions

- Construct a finite-dimensional subspace $V_h \subset V$ as follows:
- Define intervals $I_j = (x_{j-1}, x_j)$ of length $h_j = x_j - x_{j-1}$, $j = 1, 2, \ldots, (M+1)$ and set $h = \max_j h_j$.
- Let V_h be the set of functions v such that:
  - v is linear on each subinterval I_j
  - v is continuous on $[0, 1]$ and
  - $v(0) = v(1) = 0$.
- The quantity $h = \max_j h_j$ is a measure of how fine the partition is.



Continuous piecewise linear basis functions $\phi_j(x)$, $j = 1, 2, \ldots, M$



- A function $v \in V_h$ has the representation:

  $v(x) = \Sigma_{i=1}^{M} \eta_i \; \phi_i(x)$, $x \in [0, 1]$, where:

- $\eta_i = v(x_i)$ and

- $\phi_j\left(x_i\right) = \delta_{ij} \equiv \begin{cases} 1 & \text{if} \quad i = j \\ 0 & \text{if} \quad i \neq j \end{cases} \qquad i, j = 1, \ldots, M.$

- The space V_h is a linear space of dimension M with basis $\{\phi_i\}_{i=1}^{M}$.

Finite dimensional problems (M_h) and (V_h)

(M_h): Find $u_h \in V_h$ such that $F(u_h) \leq F(v) \;\; \forall\, v \in V_h$.

(V_h): Find $u_h \in V_h$ such that $(u_h', v') = (f, v) \;\; \forall\, v \in V_h$.

where:

- $u_h(x) = \Sigma_{j=1}^{M} \xi_j \phi_j(x)$, with
- $\xi_j = u_h(x_j)$ (nodal values of $u_h(x)$).

Linear system of equations $A\xi = b$

(V_h): Find $u_h \in V_h$ such that $(u_h', \phi_i') = (f, \phi_i)$, $i = 1, 2, \ldots, M$.

We finally obtain the following linear system of equations:

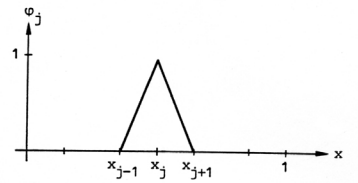
$$\Sigma_{j=1}^{M} \xi_j \, (\phi_j', \phi_i') = (f, \phi_i), \quad i = 1, 2, \ldots, M$$

where

- $A = (a_{ij})$ is the $M \times M$ <u>stiffness matrix</u> with $a_{ij} = (\phi_j', \phi_i')$
- $b = (b_i)$ is the <u>force vector</u> with: $b_i = (f, \phi_i)$ and
- $\xi = (\xi_i)$ is the <u>solution vector</u> with: $\xi_i = u_h(x_i), \quad i = 1, 2, \ldots, M$.

Properties of the stiffness matrix A

- A is symmetric, $A_{ij} = A_{ji}$, $(\phi_j', \phi_i') = (\phi_i', \phi_j')$, $i, j = 1, 2, \ldots, M$.
- A is sparse (i.e. only a few elements of A are nonzero)
  - $(\phi_j', \phi_j') = \int_{x_{j-1}}^{x_j} \frac{1}{h_j^2}\, dx + \int_{x_j}^{x_{j+1}} \frac{1}{h_{j+1}^2}\, dx$
    $= \frac{1}{h_j} + \frac{1}{h_{j+1}}, \;\; j = 1, 2, \ldots, M.$
  - $(\phi_j', \phi_{j-1}') = (\phi_{j-1,}', \phi_j') = -\int_{x_{j-1}}^{x_j} \frac{1}{h_j^2}\, dx$
    $= -\frac{1}{h_j}, \; j = 2, \ldots, M.$



  - $(\phi_i', \phi_j') = 0$ if $|i - j| > 1$.
- A is positive definite. Indeed for $\forall\, \eta \in \Re^M$ we obtain:
  - $\eta \bullet A\eta = \Sigma_{i=1}^M \Sigma_{j=1}^M \eta_i A_{ij} \eta_j = \Sigma_{i=1}^M \Sigma_{j=1}^M \eta_i (\phi_i', \phi_j') \eta_j$
    $= (\Sigma_{i=1}^M \eta_i \phi_i', \Sigma_{j=1}^M \eta_j \phi_j') = (v', v') \geq 0, \;\; v = \Sigma_{i=1}^M \eta_i \phi_i(x).$
  - Also, $\eta \bullet A\eta = 0$ only if $\eta_j = 0$, $j = 1, \ldots, M$.

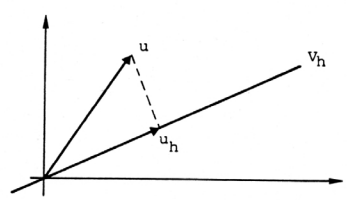
Properties of the stiffness matrix A

- Since A is a positive definite matrix, we conclude that A is non-singular.
- It follows that the system $A\xi = b$ has a unique solution.
- For the particular case of $h_j = h = \frac{1}{M+1}$, the system $A\xi = b$ becomes:

$$\frac{1}{h}\begin{bmatrix} 2 & -1 & 0 & 0 & 0 & \cdot & \cdot & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & \cdot & \cdot \\ 0 & -1 & 2 & -1 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 0 & 0 & -1 & 2 & -1 \\ 0 & \cdot & \cdot & 0 & 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \\ \cdot \\ \cdot \\ \cdot \\ \\ \\ \xi_M \end{bmatrix} = \begin{bmatrix} b_1 \\ \\ \cdot \\ \cdot \\ \cdot \\ \\ \\ b_M \end{bmatrix}$$

An error estimate for FEM for the model problem

- Let u_h be the solution of the finite element problem (V_h), i.e. $(u_h', v') = (f, v), \ \forall \ v \ \in \ V_h \quad (*)$
- Let u be the solution to problem (D), it then satisfies:
  $(u', v') = (f, v), \ \forall \ v \ \in \ V.$
  Since $V_h \subset V$, we can also write:
  $(u', v') = (f, v), \ \forall \ v \ \in \ V_h \qquad (*)$
- Subtracting the equations (*) above results in:
  $(u' - u_h', v') = 0, \ \forall \ v \ \in \ V_h$



An error estimate for the FEM for the model problem

- Let us define the norm $\|w\|$ associated with the scalar product $(.,.)$ as:

  $\|w\| = (w,w)^{\frac{1}{2}} = \left(\int_0^1 w^2 dx\right)^{\frac{1}{2}}, \ \forall\ w \in V$

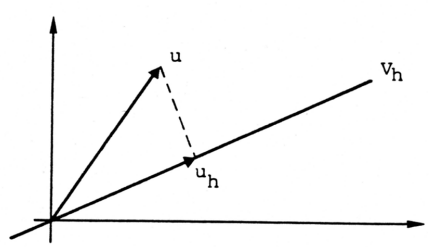
- Recall the Cauchy-Schwarz inequality:

  $|\,(v,w)\,| \leq \|v\|\ \|w\|, \ \forall\ v,w \in V$

- We can now prove that:

  Theorem: For any $v\ \epsilon\ V_h$ the following holds:

$$\|\,(u-u_h)'\,\| \leq \|\,(u-v)'\,\|$$



Theorem: For any $v \in V_h$ we have $\| (u - u_h)' \| \leq \| (u - v)' \|$

- Let $v \in V_h$ be arbitrary and set $w = u_h - v$.
- Then $w \in V_h$ and using the error condition
  $(u' - u_h', v') = 0, \ \forall \ v \in V_h$, for $v = w$, we obtain:
  $(u' - u_h', w') = 0$.
- Using the Cauchy inequality we also have:

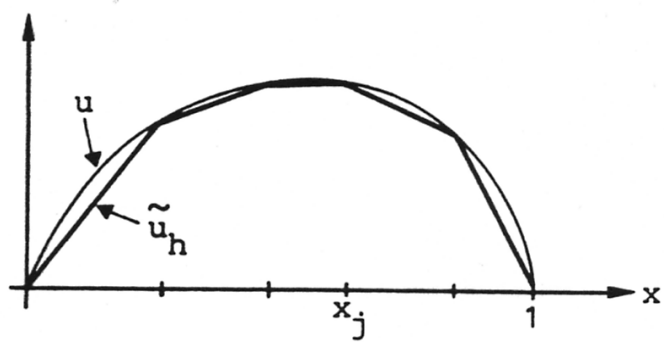
$$\begin{aligned} \| (u - u_h)' \|^2 &= \left((u - u_h)', (u - u_h)' \right) + \left((u - u_h)', w' \right) \\ &= \left((u - u_h)', (u - u_h + w)' \right) = \left((u - u_h)', (u - v)' \right) \\ &\leq \| (u - u_h)' \| \, \| (u - v)' \| \end{aligned}$$

- Dividing by $\| (u - u_h)' \|$, we obtain the required theorem.
- We conclude that $\| (u - u_h)' \|$ is as small as possible.

The finite element interpolant

- Based on the previous theorem, $\| (u - u_h)' \| \leq \| (u - v)' \|$, $\forall\ v \in V_h$, we can obtain estimates of the error $\| (u - u_h)' \|$.
- We calculate $\| (u - \tilde{u}_h)' \|$, where $\tilde{u}_h$ is the FEM interpolant of u, i.e.

$$\tilde{u}_h(x_j) = u(x_j),\ j = 0, 1, 2, \ldots, M + 1$$



Interpolation error and finite element method error

- Interpolation error:

  $$\left|u'(x) - \tilde{u}_h'(x)\right| \leq h \max_{0\leq y\leq 1}\left|u''(y)\right|,$$

  $$\left|u(x) - \tilde{u}_h(x)\right| \leq \frac{h^2}{8} \max_{0\leq y\leq 1}\left|u''(y)\right|$$

- FEM error:

  $$\left\|(u-u_h)'\right\| \leq h \max_{0\leq y\leq 1}\left|u''(y)\right|$$

  $$\left|u(x) - u_h(x)\right| \leq h \max_{0\leq y\leq 1}\left|u''(y)\right| \text{ for } 0 \leq x \leq 1 \left\{ \begin{array}{c} \text{not the best possible} \\ \text{error estimate} \end{array} \right.$$

- The error tends to zero as the max length of the subinterval I_j tends to zero (assuming that u'' is bounded on $[0,1]$).

Proof of $|u'(x) - \tilde{u}_h'(x)| \leq h \max_{0\leq y\leq 1} |u''(y)|, \ \forall \ x \in [0,1]$

1. Let $x \ \in I_{j+1} = [x_j, x_{j+1}]$, $\phi_j = \frac{x_{j+1}-x}{h_{j+1}}$, $\phi_{j+1} = \frac{x-x_j}{h_{j+1}}$.
2. $u(x_m) = u(x) + u'(x)\,(x_m - x) + \frac{1}{2}u''(\xi_m)(x_m - x)^2$, $\xi_m \in I_{j+1}$, $m = j, j+1$.
3. Multiplication of (2) with ϕ_m and addition in $m = j, j+1$, results in:
$$\Sigma_{m=j}^{j+1} \ \phi_m u(x_m) = u(x)\,\Sigma_{m=j}^{j+1}\phi_m + u'(x)\,\Sigma_{m=j}^{j+1}\phi_m(x_m - x) + \Sigma_{m=j}^{j+1}\frac{1}{2}u''(\xi_m)\phi_m(x)(x_m - x)^2$$
4. Using $\Sigma\,\phi_m x_m = x$, $\Sigma\,\phi_m = 1$ and the definition of $\tilde{u}(x)$, result in:
$$u(x) - \tilde{u}(x) = -\,\Sigma_{m=j}^{j+1}\frac{1}{2}u''(\xi_m)\phi_m(x)(x_m - x)^2$$
5. $u'(x) - \tilde{u}'(x) = -\,\Sigma_{m=j}^{j+1}\frac{1}{2}u''(\xi_m)(\phi_m'(x)(x_m - x)^2 - 2\phi_m(x)(x_m - x))$
6. Using $|u''(\xi_m)| \leq \max_{0\leq y\leq 1}|u''(y)|$, $\Sigma\,\phi_m x_m = x$, $\Sigma\,\phi_m = 1$, substituting ϕ_m' and summing in m,
$$|u'(x) - \tilde{u}'(x)| \leq \max_{0\leq y\leq 1}|u''(y)|\,\frac{1}{2}|x_{j+1} + x_j - 2x|$$
7. But $|\frac{x_{j+1}+x_j}{2} - x| \leq \frac{h_j}{2}$ and the theorem is proved.

Proof of $|u(x) - \tilde{u}_h(x)| \leq \frac{h^2}{8} \max_{0 \leq y \leq 1} |u''(y)|, \ \forall \ x \ \in \ [0,1]$

1. Recall from the previous derivation that:
$$u(x) - \tilde{u}(x) = -\Sigma_{m=j}^{j+1} \frac{1}{2} u''(\xi_m)\phi_m(x)(x_m - x)^2$$
2. With simplifications, we can write:
$$u(x) - \tilde{u}(x) = -\Sigma_{m=j}^{j+1} \frac{1}{2} u''(\xi_m)(x_m^2 + x^2 - 2x_m x)\phi_m$$
3. Using $\Sigma\, \phi_m x_m = x$ results in:
$$|u(x) - \tilde{u}(x)| \leq \frac{1}{2} \max_{0 \leq y \leq 1} |u''(y)| \, (\Sigma_{m=j}^{j+1} x_m^2 \phi_m - x^2) \qquad (*)$$
4. Let us define $E(x) = \Sigma_{m=j}^{j+1} x_m^2 \phi_m - x^2 \geq 0$. It is easy to calculate from $E'(x) = 0$ that $E(x)$ becomes maximum ($E''(\bar{x}) = -2 < 0$) at $\bar{x} = \frac{h_j}{2} + x_j$ and that $E(\bar{x}) = \frac{h_j^2}{4}$.
5. Using the above result, Equation (*) can be re-written as follows:
$$|u(x) - \tilde{u}(x)| \leq \frac{h^2}{8} \max_{0 \leq y \leq 1} |u''(y)|$$
which completes the proof.

Proof of $\| (u - u_h)' \| \leq h \max_{0 \leq y \leq 1} |u''(y)|$

1. From the main theorem we have that: $\| (u - u_h)' \| \leq \| (u - \tilde{u}_h)' \|$.
2. Using the results from the interpolation error calculations gives that:

$$\begin{aligned} \| (u - u_h)' \| &= \left(\int_0^1 (u'(z) - u_h'(z))^2 \, dz\right)^{1/2} \\ &\leq \| (u' - \tilde{u}_h') \| = \left(\int_0^1 (u'(z) - \tilde{u}_h'(z))^2 \, dz\right)^{1/2} \\ &\leq \left(\int_0^1 h^2 (\max_{0 \leq y \leq 1} |u''(y)|)^2 dz\right)^{1/2} \\ &= h \max_{0 \leq y \leq 1} |u''(y)| \end{aligned}$$

and the required result was obtained.

Proof of $|u(x) - u_h(x)| \leq h \max_{0 \leq y \leq 1} |u''(y)|, \ \forall \ x \ \in \ [0,1]$

1. Previosuly we have shown that $\| (u - u_h)' \| \leq h \max_{0 \leq y \leq 1} |u''(y)|$.
2. We can now using $u(0) = u_h(0) = 0$ write that:

$$
\begin{aligned}
|u(x) - u_h(x)| &= |\textstyle\int_0^x (u'(z) - u_h'(z)) dz| \\
&\leq \textstyle\int_0^x |u'(z) - u_h'(z)| dz \\
&\leq \textstyle\int_0^1 |(u'(z) - u_h'(z))| \ \ 1 \ dz \\
&\leq \| (u - u_h)' \| \|1\| \qquad \text{(using the Cauchy inequality)} \\
&\leq h \max_{0 \leq y \leq 1} |u''(y)| \ \text{(using (1) above)}
\end{aligned}
$$

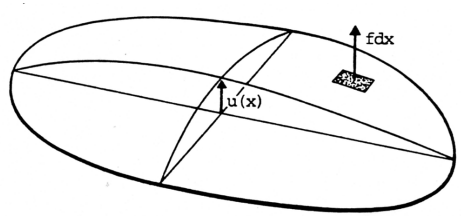
and the required result was obtained.

FEM FOR THE POISSON EQUATION

$$\boxed{\begin{aligned} -\Delta u &= f \quad \text{in } \Omega \\ u &= 0 \quad \text{on } \Gamma \end{aligned}}$$

STRONG OR DIFFERENTIAL PROBLEM (D)

- $\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2}$
- Ω is a bounded open domain in the plane $\Re^2 = \{x = (x_1, x_2) : \ x_i \ \in \ \Re\}$, with a boundary Γ.
- As an example, u can be the displacement of an elastic membrane fixed at the boundary under a transverse load f.



THE MINIMIZATION PROBLEM (M)

- Define the linear functional $F : V \longrightarrow \Re$ as:
  $F(v) = \frac{1}{2}\, a(v,v) - (f,v)$
- Find $u \in V$ such that $F(u) \quad \leq \quad F(v) \;\; \forall\, v \in V$

where:

- $V = \{v : v$ is a continuous function on Ω, $\frac{\partial v}{\partial x_1}$ and $\frac{\partial v}{\partial x_2}$ are piecewise continuous on Ω, and $v = 0$ on $\Gamma\}$
- $a\,(u,v) = \int\limits_{\Omega} \nabla u \cdot \nabla v \; dx = \int\limits_{\Omega} \left[\frac{\partial u}{\partial x_1}\frac{\partial v}{\partial x_1} + \frac{\partial u}{\partial x_2}\frac{\partial v}{\partial x_2}\right] dx$
- $(f,v) = \int\limits_{\Omega} f v dx$
- (M) corresponds to the *'Principle of Minimum Potential Energy in Mechanics'*.

THE VARIATIONAL (WEAK) PROBLEM (V)

$$\boxed{\text{Find } u \in V \text{ such that } a(u,v) = (f,v) \;\; \forall\, v \in V}$$

where:

- $V = \{v : v$ is a continuous function on Ω, $\frac{\partial v}{\partial x_1}$ and $\frac{\partial v}{\partial x_2}$ are piecewise continuous on Ω, and $v = 0$ on $\Gamma\}$
- $a\,(u,v) = \int\limits_{\Omega} \nabla u \cdot \nabla v \; dx = \int\limits_{\Omega} \left[\frac{\partial u}{\partial x_1}\frac{\partial v}{\partial x_1} + \frac{\partial u}{\partial x_2}\frac{\partial v}{\partial x_2}\right] dx$
- $(f,v) = \int\limits_{\Omega} f v dx$
- v is the so called test function
- (V) corresponds to the *'Principle of Virtual Work in Mechanics'*.

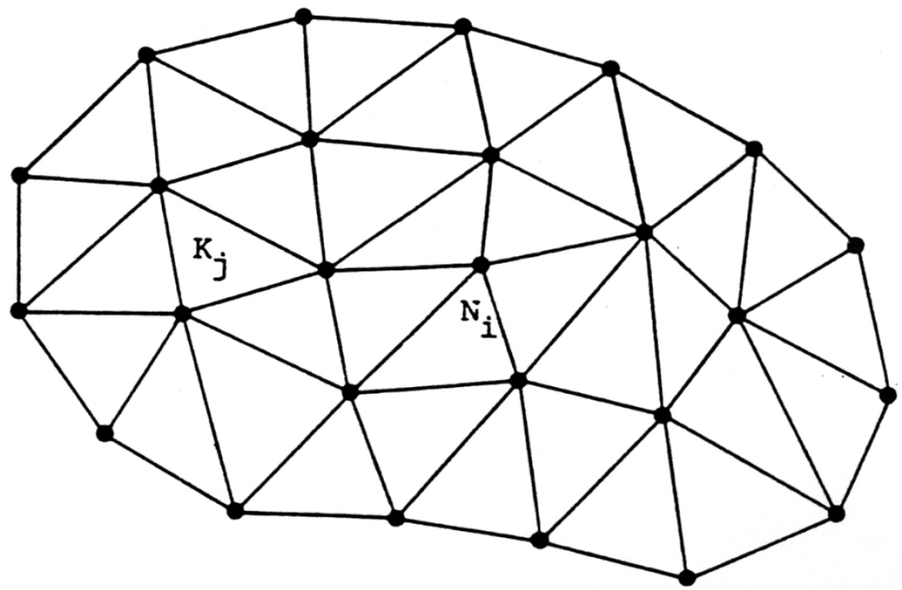
FEM for the model problem with piecewise linear functions

- Construct a finite-dimensional subspace $V_h \subset V$ as follows:
- Approximate Γ with a polygonal curve.
- Make a triangulation of Ω by subdividing Ω into a set $T_h = \{K_1, K_2, \ldots, K_m\}$ of non-overlapping triangles K_i, such that no vertex of one triangle lies on the edge of another triangle,
$$\Omega = \cup_{K \in T_h} K = K_1 \cup K_2 \ldots \cup K_m$$
- We introduce the mesh parameter h as follows: $h = \max_{K \in T_h} \text{diam}\,(K)$, where, $\text{diam}(K)$ = diameter of K = longest side of K.
- Let us define $V_h \subset V$ to be the set of functions v such that:

$$V_h = \{v : \ v \ \text{ is continuous on } \Omega, v|_K \text{ is linear for } K \in T_h, v = 0 \text{ on } \Gamma\}$$

where $v|_K$ denotes the restriction of v to K (i.e. the function defined on K agreeing with v on K).

A finite element triangulation



Continuous piecewise linear basis functions ϕ_j, $j = 1, 2, \ldots, M$

- To describe $v \in V_h$, we choose the values $v(N_i)$ at the nodes N_i, $i = 1, \ldots, M$, of T_h, but exclude the nodes on the boundary since $v = 0$ on Γ. v has then the representation:

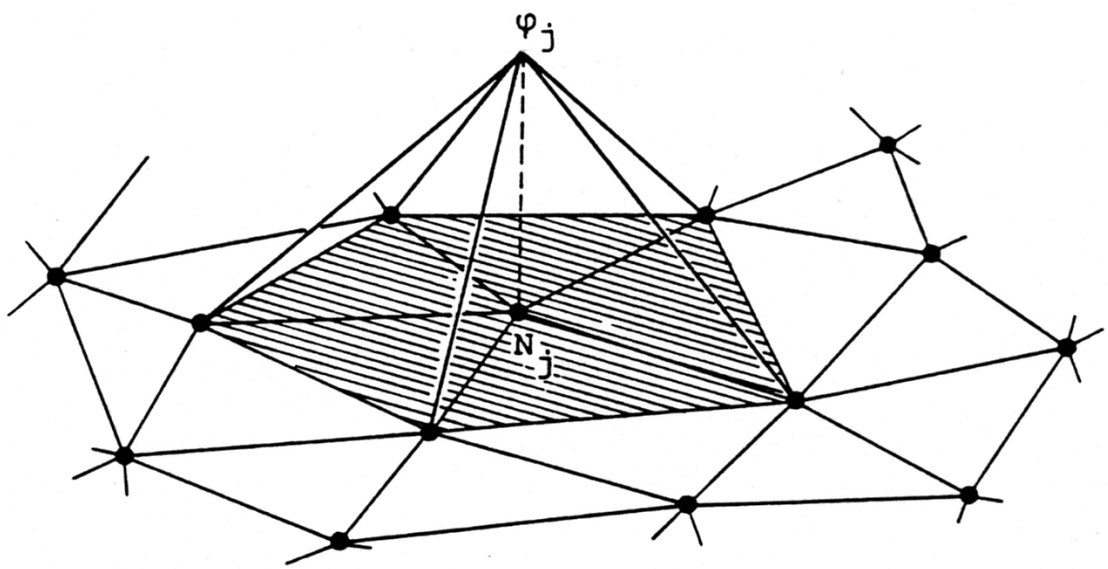
  $v(x) = \Sigma_{i=1}^{M} \eta_i \; \phi_i(x)$, $x \in \Omega \cup \Gamma$, where:

  – $\eta_i = v(N_i)$ and

  – $\phi_j\,(N_i) = \delta_{ij} \equiv \begin{cases} 1 & \text{if} \quad i = j \\ 0 & \text{if} \quad i \neq j \end{cases} \qquad i, j = 1, \ldots, M.$

- The space V_h is a linear space of dimension M with basis $\{\phi_i\}_{i=1}^{M}$.

The basis function ϕ_j



Finite dimensional problems (M_h) and (V_h)

- (M_h): Find $u_h \in V_h$ such that $F(u_h) \leq F(v) \;\; \forall\, v \in V_h$

- (V_h): Find $u_h \in V_h$ such that $a(u_h, v) = (f, v) \;\; \forall\, v \in V_h$

where:

- $u_h(x) = \Sigma_{j=1}^{M} \xi_j \phi_j(x)$, with
- $\xi_j = u_h(N_j)$ (nodal values of $u_h(x)$).

Linear system of equations $A\xi = b$

$$\boxed{(V_h): \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h, \phi_i) = (f, \phi_i),\ i = 1, 2, \ldots, M}$$

We finally obtain the following linear system of equations:

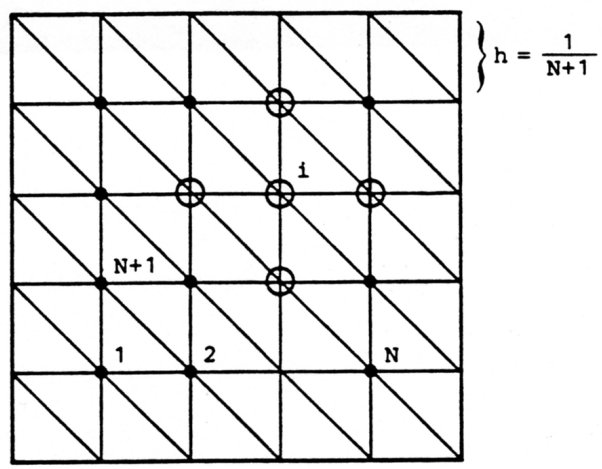
$$\boxed{\Sigma_{j=1}^{M} \xi_j \ a(\phi_j, \phi_i) = (f, \phi_i), \qquad i = 1, 2, \ldots, M}$$

where

- $A = (a_{ij})$ is the $M \times M$ <u>stiffness matrix</u> with $a_{ij} = a(\phi_j, \phi_i)$
- $b = (b_i)$ is the <u>force vector</u> with: $b_i = (f, \phi_i)$ and
- $\xi = (\xi_i)$ is the <u>solution vector</u> with: $\xi_i = u_h(N_i), \quad i = 1, 2, \ldots, M.$

Special case of uniform discretization

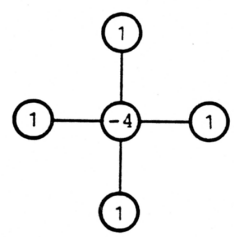
Consider Ω to be a square with side length 1 and let T_h be the uniform triangulation of Ω with the indicated node enumeration ($h = \frac{1}{N+1}$).



Special case of uniform discretization

$$\frac{1}{h}\begin{bmatrix} 4 & -1 & 0 & -1 & 0 & \cdot & \cdot & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & \cdot & \cdot \\ 0 & -1 & 4 & -1 & 0 & -1 & \cdot & \cdot \\ -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & -1 \\ \cdot & 0 & \cdot & \cdot & \cdot & \cdot & -1 & 0 \\ \cdot & \cdot & \cdot & -1 & 0 & -1 & 4 & -1 \\ 0 & \cdot & \cdot & 0 & -1 & 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \\ \cdot \\ \cdot \\ \cdot \\ \\ \\ \xi_M \end{bmatrix} = \begin{bmatrix} b_1 \\ \\ \cdot \\ \cdot \\ \cdot \\ \\ \\ b_M \end{bmatrix}$$

Note that the stiffness matrix is the same as that in the 5 point finite difference method. However, the force vector contains averages of f around each node N_i.



Interpolation error and finite element method error

- The best approximation theorem: $\|\nabla u - \nabla u_h\| \leq \|\nabla u - \nabla v\|, \ \forall \ v \ \in \ V_h,$

  $\|\nabla v\| = a\left(v, v\right)^{\frac{1}{2}} = \left(\int_\Omega |\nabla v|^2 \, dx\right)^{\frac{1}{2}} = \left\{\int_\Omega \left[\left(\frac{\partial v}{\partial x_1}\right)^2 + \left(\frac{\partial v}{\partial x_2}\right)^2\right] dx\right\}^{\frac{1}{2}}, \ \forall \ v \ \in \ V_h$

- From the above theorem we have the following FEM error estimate:

  $$\|\nabla u - \nabla u_h\| \leq \|\nabla u - \nabla \tilde{u}_h\|$$

  with $\tilde{u}_h$ the FE interpolant, i.e. $\tilde{u}_h \in V_h$ and $\tilde{u}_h\left(N_i\right) = u\left(N_i\right), i = 1, \ldots, M.$

- FEM error: If $K \ \in \ T_h$ are not allowed to become too thin, we can show:

  $$\|\nabla u - \nabla \tilde{u}_h\| \leq Ch$$

  $$\|u - u_h\| \equiv \left(\int_\Omega \left(u - u_h\right)^2 dx\right)^{\frac{1}{2}} \leq Ch^2$$

  where C is h-independent and depends only on the size of the second derivatives of u and the smallest angle of the triangles $K \ in \ T_h$.

- The error and the gradient of the error $u - u_h$ tend to zero in the norm $\| \bullet \|$ as h tends to zero (assuming that the exact solution u is sufficiently regular).

Some preliminary computational aspects

- To calculate the stiffness matrix A, assemble over element contributions:

$$a\left(\phi_i, \phi_j\right) = \sum_{K \epsilon T_h} a_K\left(\phi_i, \phi_j\right),$$

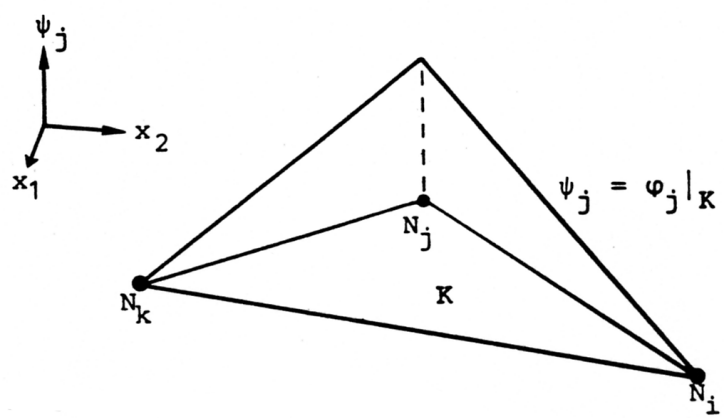
where

$$a_K\left(\phi_i, \phi_j\right) = \int_K \nabla \phi_i \cdot \nabla \phi_j dx$$

- Let N_i, N_j and N_k be the vertices of the triangle K. We define the element stiffness matrix for K as follows:

$$\begin{bmatrix} a_K\left(\phi_i, \phi_i\right) & a_K\left(\phi_i, \phi_j\right) & a_K\left(\phi_i, \phi_k\right) \\ & a_K\left(\phi_j, \phi_j\right) & a_K\left(\phi_j, \phi_k\right) \\ \text{sym} & & a_K\left(\phi_k, \phi_k\right) \end{bmatrix}$$

The basis function ψ_j associated with K.



- To compute the element stiffness matrix of the triangle K, we work with the restrictions of the basis functions ϕ_j on K, $\psi_j = \phi_j|_K$.
- We call ψ_i, ψ_j and ψ_k the basis functions on the triangle K.
- If $w \in V_h$, its representation on K is the following:

$$w(x) = w(N_i)\,\psi_i(x) + w(N_j)\,\psi_j(x) + w(N_k)\,\psi_k(x)$$

The Hilbert spaces $L_2(\Omega)$, $H^1(\Omega)$ and $H_0^1(\Omega)$

- While working with variational forms of BVP's for PDE's, it is natural and useful to work with function spaces V that are slightly larger than the spaces of continuous functions with piecewise continuous derivatives used previously.
- We will endow the space V with the scalar product related to the BVP. The space V will be a Hilbert space.

Linear and bi-linear forms

- L is a *linear form* on V if $L: \ V \longrightarrow \Re$, i. e., $L(v) \in \Re$ for $v \in V$, and L is linear, i. e., for all $v, w \in V$ and $\beta, \theta \in \Re$
$$L(\beta v + \theta w) = \beta L(v) + \theta L(w)$$
- We say that $a(.,.)$ is a *bi-linear form* on $V \times V$ if $a: V \times V \longrightarrow \Re$, i. e., $a(v, w) \in \Re$ for $v, w \in V$, and a is linear in each argument, i. e., for all $u, v, w \in V$ and $\beta, \theta \in \Re$, the following holds:
$$a(u, \beta v + \theta w) = \beta a(u, v) + \theta a(u, w)$$
$$a(\beta u + \theta v, w) = \beta a(u, w) + \theta a(v, w)$$
- The bilinear form $a(.,.)$ on $V \times V$ is said to be *symmetric* if:
$$a(v, w) = a(w, v), \qquad \forall \ v, w \in V$$

Inner product and norm related to the BVP

- A symmetric bilinear form $a(.,.)$ on $V \times V$ is said to be a *scalar product* on V if:
$$a(v,v) > 0 \qquad \forall\, v \in V, v \neq 0$$
- The *norm* $\|\cdot\|_a$ associated with a scalar product $a(.,.)$ is defined by
$$\|v\|_a = (a(v,v))^{\frac{1}{2}}, \qquad \forall\, v \in V \qquad \text{“energy norm”}$$
- Further, if $< .,. >$ is a scalar product with corresponding norm $\|\cdot\|$, then we have the *Cauchy-Schwarz inequality:*
$$|< v,w >| \leq \|v\|\,\|w\|$$

Hilbert spaces

- Let V be a linear space with a scalar product and corresponding norm $\|\cdot\|$.
- A sequence v_1, v_2, ..., of elements v_i in the space V with norm $\|\cdot\|$ is said to be a Cauchy sequence if for all $\epsilon > 0$ there is a natural number N such that
$$\|v_i - v_j\| < \epsilon \;\; \forall\, i,\, j > N$$
- v_i converges to v if
$$\|v - v_i\| \longrightarrow 0 \text{ as } i \longrightarrow \infty$$
- The space V is said to be a *Hilbert space* if V is complete, i. e., if every *Cauchy sequence* with respect to $\|\cdot\|$ is convergent.

Hilbert spaces for variational problems

- If $I = (a, b)$, define *the space of square integrable functions on* I as:
$$Ł_2(I) = \left\{ v : \ v \text{ is defined on } I \text{ and } \int_I v^2 dx < \infty \right\}$$
- The space $L_2(I)$ is a Hilbert space with the scalar product:
$$(v, w) = \int_I vw dx$$
- The corresponding norm is defined as: $\|v\|_{L_2(I)} = \left(\int_I v^2 dx \right)^{\frac{1}{2}} = (v, v)^{\frac{1}{2}}$
- By Cauchy's inequality, we have:
$$|(v, w)| \leq \|v\|_{L_2(I)} \|w\|_{L_2(I)} \qquad (*)$$
- We conclude from Equ. (*) that (v, w) exists, if v and $w \in L_2(I)$.
- Think of a typical $v \in L_2(I)$ as a piecewise continuous function, possibly unbounded, such that $\int_I v^2 dx < \infty$

  Example: $v = x^{-\beta} \in L_2(I)$, $I = (0, 1)$ for $\beta < \frac{1}{2}$.

The Sobolev $H^1(I)$ space

- We define the space $H^1(I)$ as follows:
$$H^1(I) = \{v : \ v \text{ and } v' \ \in L_2(I) \ \}$$
- We equip $H^1(I)$ with the following scalar product:
$$(v, w)_{H^1(I)} = \int_I (vw + v'w')\, dx$$
- The corresponding norm is:
$$\|v\|_{H^1(I)} = \left(\int_I \left[v^2 + (v')^2 \right] dx \right)^{\frac{1}{2}}$$

The Sobolev $H_0^1(I)$ space

- Recall our model problem: $-u''(x) = f(x)$, $x \in (0,1)$, $u(0) = u(1) = 0$.
  For this problem, we need to use the space $H_0^1(I)$ defined as:
  $$H_0^1(I) = \left\{ v \in H^1(I) : v(0) = v(1) = 0 \right\}$$
  with the same scalar product and norm as for $H^1(I)$.
- The weak form for the model problem is then written as follows:
  Find $u(x) \in H_0^1(I)$ such that: $(u', v') = (f, v)$, $\forall\, v \in H_0^1(I)$,
  where $(u', v') = \int_I u'\, v'\, dx$.
- The space $H_0^1(I)$ is the largest space for which the variational problem is meaningful.
- The formulation in $H_0^1(I)$ is preferable to that in V in that the basic error estimate for the FEM is an estimate in the H^1 norm.

The spaces $L_2(\Omega)$, $H^1(\Omega)$ and $H_0^1(\Omega)$, $\Omega \subset \Re^d$, $d = 2, 3$

- $L_2(\Omega) = \left\{ v : v \text{ is defined on } \Omega \text{ and } \int_\Omega v^2 dx < \infty \right\}$ with
  - $(v, w) = \int_\Omega vw dx$
  - $\|v\|_{L_2(\Omega)} = \left(\int_\Omega v^2 dx \right)^{\frac{1}{2}}$
- $H^1(\Omega) = \left\{ v \in L_2(\Omega) : \frac{\partial v}{\partial x_i} \in L_2(\Omega), i = 1, \ldots, d \right\}$ with
  - $(v, w)_{H^1(\Omega)} = \int_\Omega \left[vw + \nabla v \cdot \nabla w \right] dx$
  - $\|v\|_{H^1(\Omega)} = \left(\int_\Omega \left[v^2 + |\nabla v|^2 \right] dx \right)^{\frac{1}{2}}$
- We also define:
  $H_0^1(\Omega) = \left\{ v \in H^1(\Omega) : v = 0 \text{ on } \Gamma \right\}$
  and the same scalar product and norm as $H^1(\Omega)$.

An example problem in $\Omega \subset \Re^d,\ d = 2, 3$

- Let us consider the following BVP:

  (D) Find u such: $-\Delta u = f$ in Ω, $u = 0$ on Γ.

- The variational form is given as follows:

  (V) Find $u \in H_0^1(\Omega)$ such that $a(u, v) = (f, v) \quad \forall\, v \in H_0^1(\Omega)$

  or equivalently

- (M) Find $u \in H_0^1(\Omega)$ such that $F(u) \leq F(v)\ \forall\, v \in H_0^1(\Omega)$

  - $F(v) = \frac{1}{2} a(v, v) - (f, v)$,
  - $a(u, v) = \int_\Omega \nabla u \cdot \nabla v dx$
  - $(f, v) = \int_\Omega f v dx$

- It may be difficult to prove the existence of classical solutions of (D).

- However, the existence of solutions of (V) (weak solutions of (D)) may be easier.

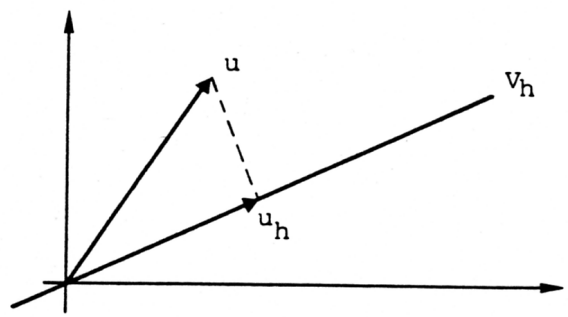
A finite element approximation in $V_h \subset H_0^1$

- $$\begin{aligned} -\Delta u + u &= f \qquad \text{in } \Omega, \\ u &= 0 \qquad \text{on } \Gamma \end{aligned}$$
- Find $u \in H_0^1$ such that:
$$< u, v >= (f, v)\,, \ \forall\, v \in H_0^1(\Omega) \qquad ,$$
where:
$< u, v >= \int_\Omega [\nabla u \cdot \nabla v + uv]\, dx$ is the scalar product in $H_0^1(\Omega)$.
- Find $u_h \in V_h \subset H_0^1(\Omega)$ such that: $< u_h, v >= (f, v)\ \forall\, v \in V_h$

A geometric interpretation of the FEM in H_0^1

- We can obtain: $< u - u_h, v >= 0, \quad \forall\, v \;\in\; V_h$, i.e. the error $u - u_h$ is orthogonal to V_h with respect to $< ., . >$.
- The finite element solution u on V_h is the *projection* with respect to $< ., . >$ of the exact solution u in V_h, i.e. u_h is the element in V_h closest to u with respect to the $H_0^1(\Omega)$ norm or in other words:

$$\|u - u_h\|_{H^1(\Omega)} \leq \|u - v\|_{H^1(\Omega)} \qquad \forall\, v \;\in\; V_h$$



The Neumann problem
Natural and essential boundary conditions

- Consider the following Newmann problem (D):

$$\begin{aligned} -\Delta u + u &= f \qquad \text{in } \Omega \\ \frac{\partial u}{\partial n} &= g \qquad \text{on } \Gamma \end{aligned}$$

- The variational formulation (V) becomes:

  Find $u \in H^1(\Omega)$ such that,

$$a(u,v) = (f,v) + < g, v > \quad \forall\, v \in H^1(\Omega)$$

  where

$$a(u,v) = \int_\Omega [\nabla u \cdot \nabla v + uv]\, dx, (f,v) = \int_\Omega fv dx, \; < g, v >= \int_\Gamma gv ds$$

- The Newmann condition $\frac{\partial u}{\partial n} = g$ does not appear explicitly in (V).
- The solution u of (V) is only required to belong to $H^1(\Omega)$ and is not explicitly required to satisfy the Newmann condition.

The Neumann problem

- Let T_h be a triangulation of Ω and let
  $V_h = \{v : v \text{ is continuous on } \Omega,\ v|_K \text{ is linear } \forall\, K \in T_h\} \subset H^1(\Omega)$
- As parameters to describe the functions in V_h, we choose the values at the nodes <u>including</u> the nodes on the boundary Γ.
- The finite element method becomes: Find $u_h \in V_h$ such that
  $$a(u_h, v) = (f, v) + < g, v > \qquad \forall\, v \in V_h$$
- The problem has a unique solution u_h that can be determined by solving a symmetric, positive definite linear system of equations. The following error estimate is obtained:
  $$\|u - u_h\|_{H^1(\Omega)} \leq \|u - v\|_{H^1(\Omega)} \quad \forall\, v \in V_h$$
  and hence as shown earlier
  $$\|u - u_h\|_{H^1(\Omega)} \leq \text{Ch}, \qquad \text{(if } u \text{ is regular enough)}$$
- The solution u_h will satisfy the Newmann condition approximately.

Show that a solution $u \in H^1(\Omega)$ of (V) is also a solution of (D)

- Using Green's theorem, we find that if u is sufficiently regular
  $(f, v) + < g, v >= a(u, v) = \int_\Gamma \frac{\partial u}{\partial n} v ds + \int_\Omega (-\Delta u + u) v dx$
  or rearranging terms:
  $\int_\Omega (-\Delta u + u - f) v dx + \int_\Gamma (\frac{\partial u}{\partial n} - g) v ds = 0, \forall\ v \in H^1(\Omega)$
- Consider the above equation for the functions v that vanish on the boundary. We then conclude that: $-\Delta u + u - f = 0$, in Ω.
- We thus have that: $\int_\Gamma (\frac{\partial u}{\partial n} - g) v ds = 0, \forall\ v \in H^1(\Omega)$.
  By varying v over $H^1(\Omega)$ (which means that v will vary freely on Γ), we finally get: $\frac{\partial u}{\partial n} = g$, on Γ.

Programming remarks

- Consider the Newmann problem examined earlier:

$$-\Delta u + u = f \qquad \text{in } \Omega$$
$$\frac{\partial u}{\partial n} = g \qquad \text{on } \Gamma$$

- Let $T_h = \{K\}$ be a triangulation of $\Omega \subset \Re^2$ with boundary Γ and let V_h be the corresponding space of continuous piecewise linear functions.
- Let $N_i, i = 1, 2, \ldots, M$, denote the nodes of T_h and $\phi_1, \ldots, \phi_M$ the natural base for V_h with $\phi_i(N_j) = \delta_{ij}$.
- We want to find the solution $\xi \in \Re^M$ of the linear system: $A\xi = b$, where $A = (a_{ij}), b = (b_1, \ldots, b_M)$ and

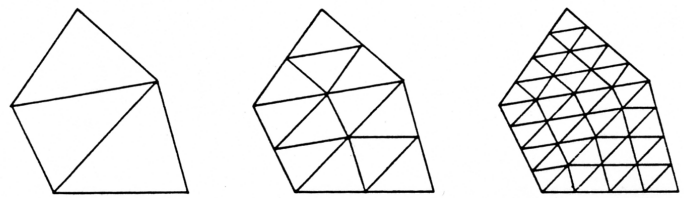
$$a_{ij} = \sum_{K \epsilon T_h} a_{ij}^K, \quad b_i = \sum_{K \epsilon T_h} b_i^K$$
$$a_{ij}^K = \int_K \left\{ \nabla \phi_i \cdot \nabla \phi_j + \phi_i \phi_j \right\} dx, \quad b_i^K = \int_K f \phi_i dx + \int_{K \cap \Gamma} g \phi_i ds$$

Main tasks in finite element programming

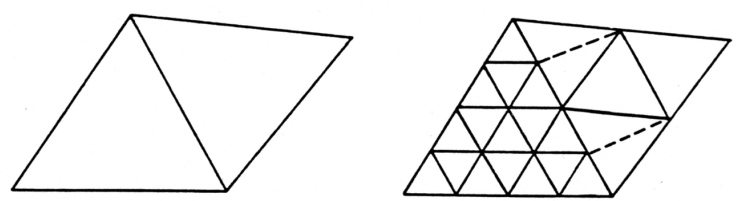
- Input the data f, g, Ω and the coefficients of the equation.
- Construct and represent the triangulation T_h.
- Compute the element stiffness matrices a^K and element loads b^K.
- Assemble the global matrix A and load vector b.
- Solve the system of equations $A\xi = b$.
- Present the results (post-processing, visualization, etc.).

Construction of the triangulation T_h

- Successive refinement by connecting the midpoints of each side (this leads to *quasi-uniform meshes* where the triangles have essentially the same size in all parts of Ω):



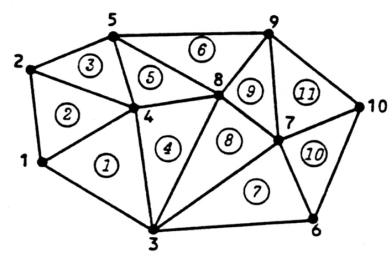
- In general, one needs smaller triangles in regions where the exact solution varies quickly or where certain derivatives of the exact solution are large (*adaptive methods*).



Representation of the triangulation T_h

- Let N_i, $i = 1, 2 \ldots, M$ and K_n, $n = 1, \ldots, N$ be enumerations of the nodes and triangles of T_h, respectively.
- Then T_h is defined using two arrays:
  - $Z(j, i)$, $j = 1, 2$ are the coordinates of node N_i.
  - $T(j, n)$, $j = 1, 2, 3$ are the number of the vertices of triangle K_n.

$$T = \begin{bmatrix} 1 & 1 & 2 & 3 & 4 & 5 & 3 & 3 & 8 & 7 & 7 \\ 3 & 4 & 4 & 8 & 8 & 8 & 6 & 7 & 7 & 6 & 10 \\ 4 & 2 & 5 & 4 & 5 & 9 & 7 & 8 & 9 & 10 & 9 \end{bmatrix}$$



- When using the Gaussian elimination to solve $A\xi = b$, the nodes need to be enumerated such that the bandwidth of A is minimal.

Computation of the element stiffness matrices

- We know that $a_{ij}^K \not\equiv 0$ only if both N_i and N_j are nodes of K. Let $K_n \in T_h$.
- $T(\alpha, n)$, $\alpha = 1, 2, 3$ are the vertices of K_n, whereas $Z(i, T(\alpha, n))$, $\alpha = 1, 2, 3$, $i = 1, 2$ are the x_i coordinates of these vertices.
- The element stiffness matrix $A^{(n)} = (a_{\alpha\beta}^n)$, α, β = 1,2,3, for element K_n is given as follows:

$$a_{\alpha\beta}^n = \int_{K_n} \left[\nabla\psi_\alpha \cdot \nabla\psi_\beta + \psi_\alpha\psi_\beta\right] dx$$

where

ψ_α is the linear function on K_n that takes the following values:

$$\psi_\alpha\left(N_{T(\beta,n)}\right) = \begin{cases} 1 \text{ if } \alpha = \beta \\ 0 \text{ if } \alpha \neq \beta \quad \alpha, \beta = 1, 2, 3 \end{cases}$$

- We also compute:

$$b_\alpha^n = \int_{K_n} f\psi_\alpha dx + \int_{\Gamma\cap K_n} g\psi_\alpha dx, \qquad \alpha = 1, 2, 3$$

Assembly of global stiffness matrix

- Set $A(i,j) = 0, \qquad b(i) = 0, \quad i,j = 1,\ldots,M.$
- For $n = 1,\ldots,N$, fetch $A^{(n)} = \left(a^n_{\alpha\beta}\right)$ and $b^{(n)} = \left(b^n_\alpha\right)$ from scratch file and for $\alpha,\beta = 1,2,3$ set:

$$A\left(T(\alpha,n), T(\beta,n)\right) = A\left(T(\alpha,n), T(\beta,n)\right) + a^n_{\alpha\beta},$$

$$b\left(T(\alpha,n)\right) = b\left(T(\alpha,n)\right) + b^n_\alpha$$

Solution of the linear system $A\xi = b$

- If Gaussian elimination is to be used, A is stored as a band matrix with an arrangement of the node numbering that leads to a minimum bandwidth.
- If an iterative method is used, only the nonzero elements of A are stored.
- Once the stiffness matrix A for the Newmann problem has been determined, we can easily derive the system $\bar{A}\xi = \bar{b}$ corresponding to any other type of boundary conditions.
- For example, if $u = u_o$ on Γ_1, we can obtain the corresponding system $\bar{A}\xi = \bar{b}$ by deleting the rows of A corresponding to the nodes on Γ_1 and by entering the values of ξ given by the Dirichlet boundary condition.